Xuyang Wang

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SUMMARY

I am a master's student in Data Science at Lehigh University, with a background in machine learning and data analytics. My research interests include **autonomous AI research**, **LLM-based agents**, and **recursive self-improvement (RSI)**. I explore verifier-guided self-improvement and adaptive research agents through automated scientific paper replication.

EDUCATION

Lehigh University **Jun 2025 - Dec 2026 (Expected)**
Master of Science in Data Science
Coursework: Accelerated Computing for ML; Deep and Generative Learning; Data Mining; Vibe Research for Scientific Discovery

Lanzhou Jiaotong University **Sep 2021 - Jun 2025**
Bachelor of Engineering in Data Science and Big Data Technology
Coursework: Data Structures & Algorithms; Database Systems; Rapid Data Analysis; Data Visualization

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++ **Data:** Pandas, NumPy, MongoDB, MySQL, ETL/Data Pipelines
ML: scikit-learn, TensorFlow/Keras, Keras Tuner, PyTorch, Hugging Face Diffusers, scikit-surprise
Visualization & Tools: Power BI, Plotly, Streamlit, IBM SPSS Modeler, Git, Linux

WORK EXPERIENCE

Pan-China Certified Public Accountants LLP **Jul 2026 - Aug 2026**
IT Audit Intern *Hangzhou, China*
- Reconciled e-commerce orders and logistics data across systems to identify OMS and shipment discrepancies; classified exception causes and flagged transactions for client clarification or audit follow-up.
- Cleaned audit data, updated analysis scripts, and performed sample testing across three e-commerce platforms.

QY (Qianyuan) Capital Co., Ltd. **Jan 2024 - Mar 2024**
Quantitative Research Intern *Shenzhen, China*
- Screened 15 candidate factors using 12 months of financial data and selected 5 for strategy review; applied a genetic algorithm over 3 iterations to prioritize factor combinations.
- Analyzed factor contributions to returns and risk to support portfolio optimization and risk-control discussions.

Capgemini Consulting (China) Co., Ltd. **Jul 2023 - Aug 2023**
Consulting Assistant Intern (Data) *Shanghai, China*
- Collected 40% of the new-energy vehicle review dataset across 12 websites, adapting Python crawlers to 7 page structures.
- Maintained MongoDB collections and 2 data-transfer interfaces; encoded three-class sentiment labels and aligned data schemas with the algorithm team to prepare NLP training data.

RELEVANT PROJECTS

Olist Business Intelligence Agent **Ongoing**
- **Skill orchestration:** Built a natural-language analytics agent with nine skills for planning, schema discovery, guarded SQL, validation, visualization, and synthesis, delivering Streamlit/Plotly reports over Olist e-commerce data.
- **Evidence-gated validation:** Encoded 31 source-data rules and policies, plus query-level reconciliation and coverage checks; used Candidate–Validate–Commit to block results with failed or missing required evidence.
- **Persistent conversation memory:** Combined SQLite history with retrieval of validated analyses and query-filter inheritance; a live API test retained region and order-status filters across eight preseeded intervening turns.
- **Checkpoint and resume:** Implemented journals and checkpoints; on a 10,000-order sample, independently reconciled order/GMV totals with CSV data and verified that resuming five committed tasks required no new API calls.

Diffusion-Based Image Stylization System **Feb 2026 - May 2026**
- Built a Stable Diffusion img2img system with PyTorch and Hugging Face Diffusers to stylize uploaded images using text prompts and adjustable generation strength; exposed these controls through Gradio and Streamlit interfaces.
- Compared diffusion outputs with CNN-based style transfer using SSIM for structural similarity and CLIP Score for semantic alignment, presenting image results and parameter trade-offs through the interfaces.

Bitcoin Price Analysis and Prediction Based on LSTM **Feb 2025 - Jun 2025**
- Prepared 3,413 BTC-USD records with forward filling, MinMax scaling, and 60-day input sequences; engineered return, rolling-volatility, and volume features for forecasting and market analysis.
- Identified four market states with PCA and K-Means, and detected unusual price and volume behavior using Z-score, IQR, and Isolation Forest; combined these signals with forecasts in a Plotly prototype with rule-based risk alerts.
- Tuned a Keras LSTM with Keras Tuner, reducing test RMSE by **28.5%**, from **1,904.98 to 1,361.78**.

Machine Learning in Recommender Systems **Jan 2023 - Apr 2023**
- Converted Amazon Sports & Outdoors ratings into user-item interactions, retaining users with at least five ratings; trained a 100-factor Surprise SVD model and applied PCA to reduce the user-item latent representations.
- Generated Top-5 personalized recommendations from similarity scores in the reduced feature space and evaluated SVD/PCA configurations using test-set RMSE.